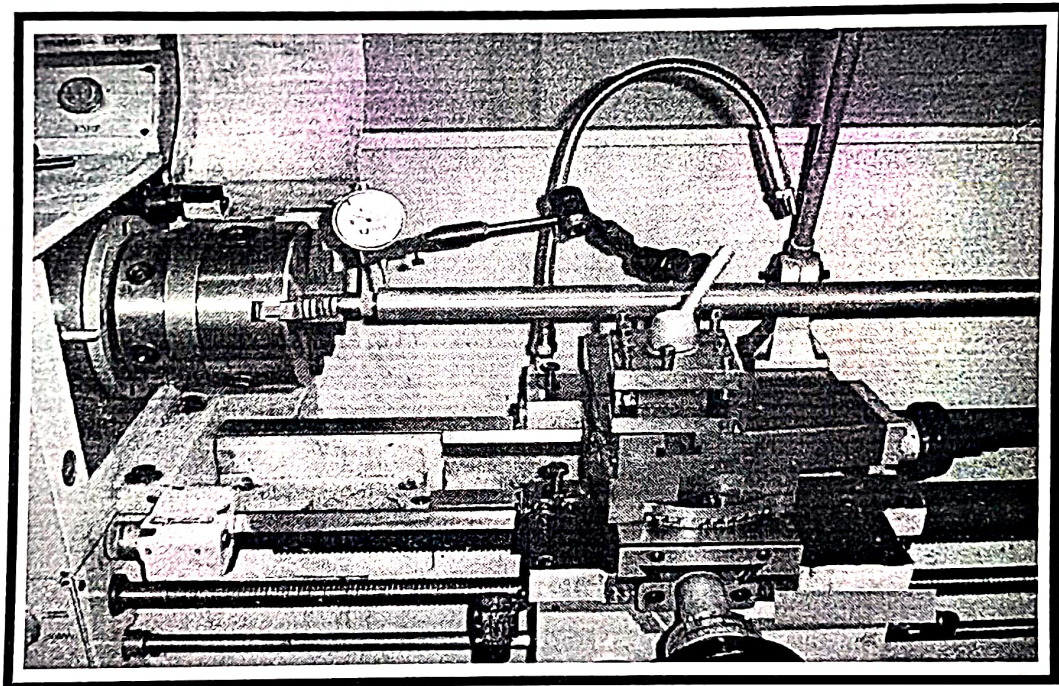


MACHINE TOOL ALIGNMENT TESTS



MACHINE TOOL TESTING

INTRODUCTION:

The surface components produced by machining processes are mostly by generation. As a result, the quality of surface produced depends upon the accuracy of the various movements of the machine tool concerned. It therefore becomes important to know the capability of the machine tool by evaluating the accuracy of the various mechanisms that are directly responsible for generating the surface. For this purpose a large variety of tests have been designed.

MEASURING INSTRUMENTS USED FOR TESTING:

The accuracy of the machine tools employed should be higher than the accuracy of the components that it produces. Similarly the quality of the measuring equipment used for machine tool testing should be commensurate with the quality expected from such testing. A few commonly used equipments are

- Dial Indicators
- Test mandrels
- Straight edges
- Spirit levels

TEST PROCEDURES:

The major tests that are conducted on machine tool are:

- Testing the quality of the slide ways and the locating surfaces
- Testing the accuracy of the main spindle and its alignment with respect to other parts of the machine tool.
- Testing the accuracy of the parts produced by the machine tool.

ACCEPTANCE TESTS

LATHE MACHINE

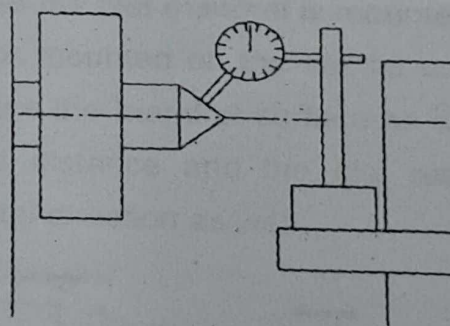
Tests that can be conducted on Lathe machine:

1. **Quality of slide ways:** To test the quality of the slide ways it is necessary to mount the dial indicator on a good datum surface. Then the plunger is moved along the longitudinal direction of the slide ways which provides an indication of the undulations present on the surface of the slide ways.

2. **Accuracy of the spindle:**

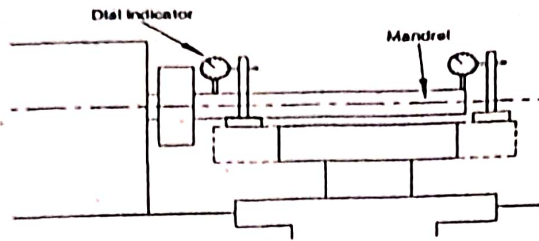
These tests are related to the true running of the spindle and the centre located in the spindle along with the alignment, parallelism and perpendicularity of the spindle with the other axes of the concerned machine tool.

True running of the centre: The live centre may be loaded into the lathe spindle and a dial indicator mounted as shown in fig. This test is required only for machines where the work piece is held between centres. The readings of the dial indicator are taken while rotating the spindle through full rotation.



*True running of center of spindle
+ w/p.*

True running of the spindle: the taper shank of the test mandrel of about 300 mm length is mounted into the spindle as shown in fig. The plunger of the dial indicator rests on the cylindrical surface of the mandrel. The spindle is rotated slowly and the readings of the dial indicator are noted. The deviation should normally be less than 0.01mm. The test is to be repeated with the dial indicator positioned close to the spindle bore as well as at the extreme end of the test mandrel.

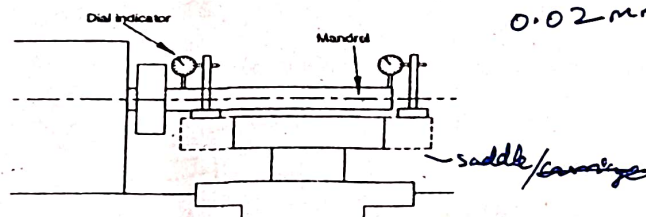


Test set-up for true running of the spindle of a lathe

Squareness of the face: this test is used to measure the squareness of the shoulder face with reference to the spindle axis. The plunger of the dial indicator rests on the extreme radial position of the shoulder face and the reading is taken. It is repeated by slowly rotating the spindle till the dial indicator comes to a point that is diametrically opposite to the reading taken earlier.

3. Alignment tests:

Parallelism and perpendicularity: Parallelism and perpendicularity between two axes or two surfaces is normally measured in two planes, horizontal and vertical. For this purpose the test mandrel is mounted in the spindle as shown in fig. with dial indicator mounted on the saddle or carriage. The plunger of the dial indicator touches the mandrel surface as shown in fig. the saddle is moved for a specified distance and the dial reading noted. The test is repeated in the horizontal direction as well.

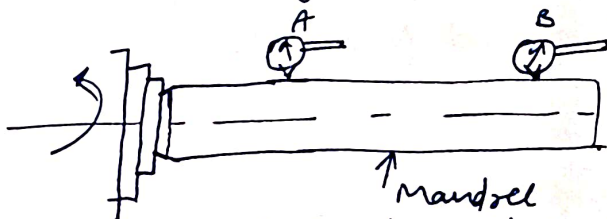


Test set-up for parallelism between the spindle axis and the slideways in a lathe

Permissible error
0.02 mm

True Running of Taper Socket⁴² in Main Spindle

Machine Tool Alignment Tests

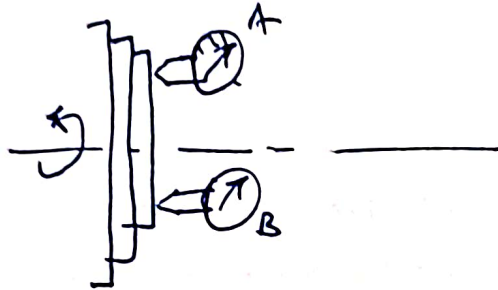


Perm. error for A = 0.01 mm
for B = 0.02 mm

Axis of the tapered socket into the main spindle is required to be co-centric. Other job produced will be eccentric or tapered.

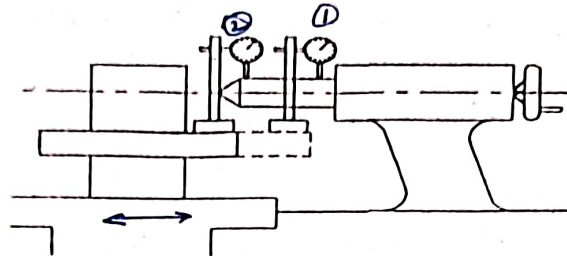
Take reading at two positions.

Axial slip of main Spindle
Squareness of face



0.01 — 0.02 mm

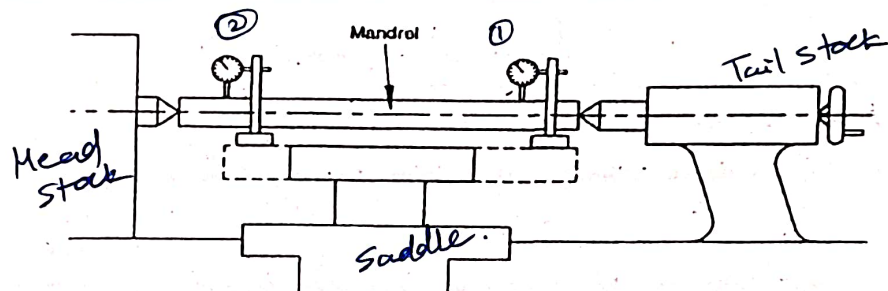
Parallelism between the outside diameter of the tail stock sleeve and the slide ways as shown in fig.



Test set-up for the parallelism of the tail stock sleeve

True cylindrical surface of job may not be achieved.

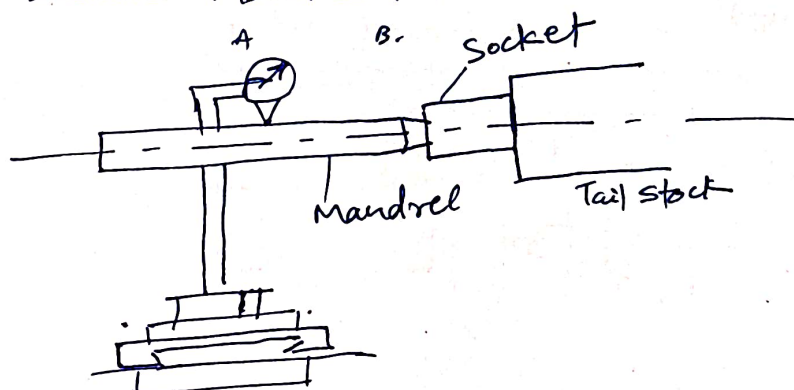
Parallelism between the line of centres and the slide ways shown in fig.



Test set-up for parallelism of the line of centres in a lathe

0.02 mm

Parallelism of Tail stock sleeve taper socket to saddle movements.



Axial slip of the Lead screw ⁴³ Machine Tool Alignment Tests

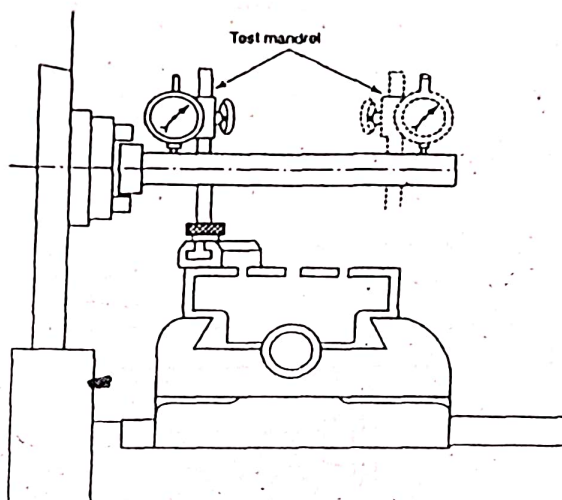
— Collar must be exactly $\perp$ to axis of screw.

MILLING MACHINE:

The following tests can be conducted:

1. True running of the spindle:

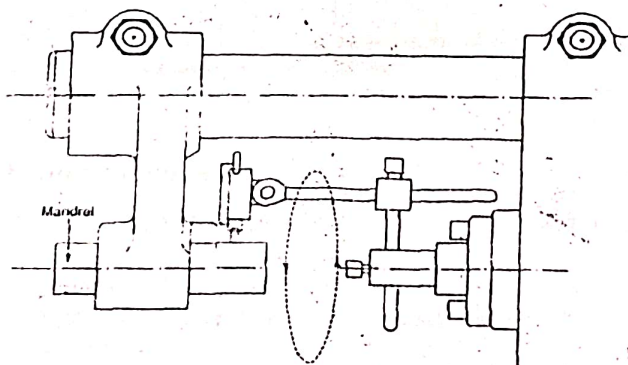
Table
Head
Vise + fixture } Alignment



Test set-up for true running of the spindle of a milling machine

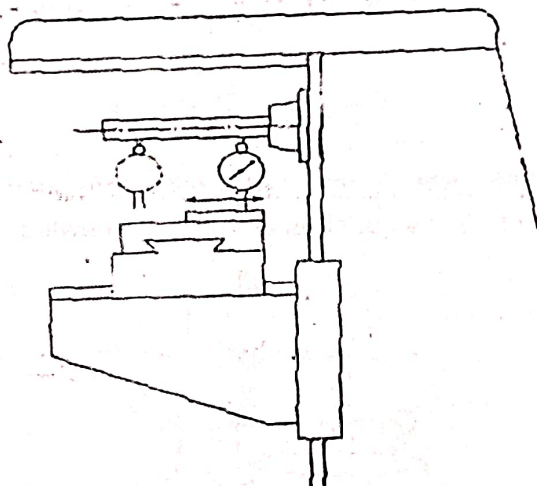
2. Spindle alignment:

In this test dial indicator is mounted on one of the surfaces whose alignment is to be tested with another surface. In case of a horizontal milling machine the testing of the alignment between the spindle and the over arm support can be done as shown in fig.



The dial indicator is mounted on the spindle while a test mandrel is mounted in the over arm support with the plunger of the dial indicator resting on the cylindrical surface of the test mandrel. The spindle is rotated and readings are taken when it is at different positions on the periphery of the test mandrel. The test may be conducted at two extreme ends of the mandrel.

Parallelism between the table and the spindle axis shown in fig. A test mandrel 300mm long is mounted in the spindle axis and the dial indicator is mounted on the table. The reading of the dial indicator is taken at the two extreme positions with out the table movement.



Test set-up for parallelism between the table and the spindle axis in a horizontal milling machine

Other tests that can be conducted are:

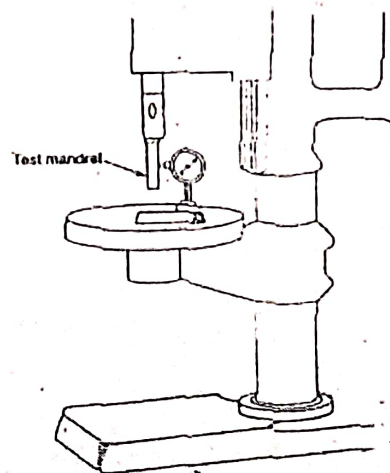
- Parallelism between the spindle axis and the transverse movement of the table.
- Perpendicularity between the spindle and the vertical column ways.

Axial Slip of spindle # Eccentricity of external dia.

RADIAL DRILLING MACHINE:

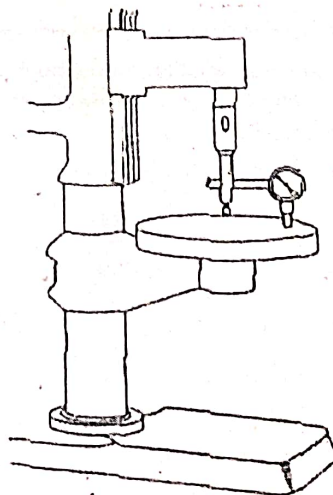
Tests that can be conducted on Drilling Machine are:

1. True running of the spindle.

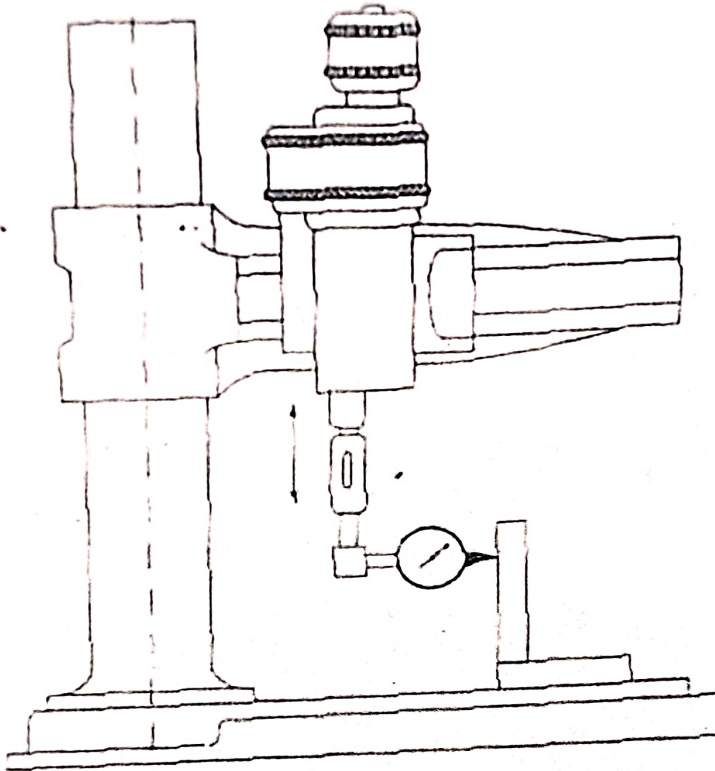


Test set-up for true running of the spindle of a radial drilling machine

2. Perpendicularity between the spindle and the base plate.



3. Perpendicularity between the feed movement and the base plate.



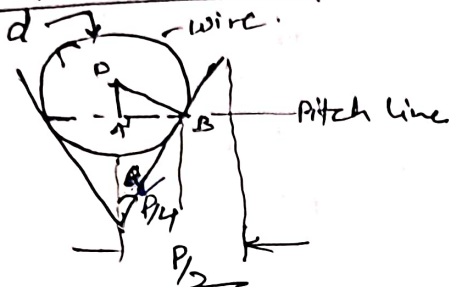
Test set-up for the perpendicularity between the feed movement and the base plate of a radial drilling machine

Screw Thread Measurement

① Pitch P — by thread gauge
screw pitch gauge

② Flank angle

③ Best wire size.



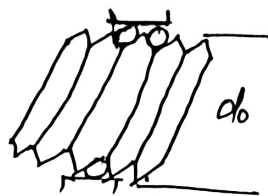
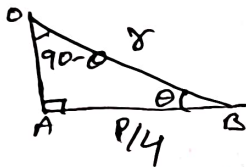
$$AB = P/4$$

$$\angle OBA = 90 - (90 - \theta) = \theta/2$$

$$\angle AOB = 90 - \theta/2$$

$$\sin(90 - \theta) = \frac{AB}{OB} \Rightarrow \cos \theta = \frac{P/4}{r}$$

$$\theta = \frac{P}{4} \sec \theta/2 \Rightarrow d = \frac{P}{2} \sec \theta/2$$



pitch measuring mic
Tool maker's microscope



Flank angle

$$\sin \theta/2 = \frac{O_2 A}{O_1 O_2}$$

$$= \frac{(d_2 - d_1)/2}{\left(\frac{M_2 - d_2}{2}\right) + \left(\frac{M_1 - d_1}{2}\right)}$$

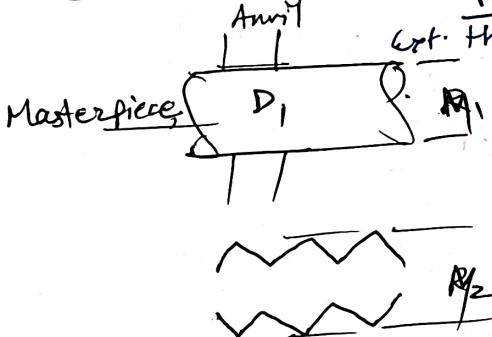
$$\sin \theta/2 = \frac{d_2 - d_1}{(M_2 - M_1) - (d_2 - d_1)}$$

for metric thread $\theta = 60^\circ$

$$d = 0.5774p$$

③ Minor diameter — by point micrometer $d_1 = D \pm (R_2 - R_1)$
 $D = \text{length slip gauge} + 2 \times \text{dia of roller}$

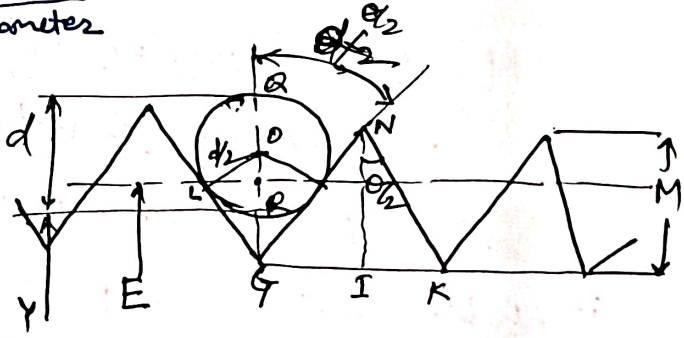
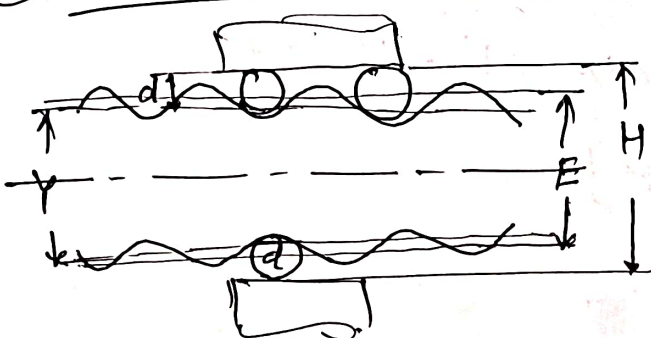
④ Major diameter (Nominal diameter) — by floating carriage micrometer



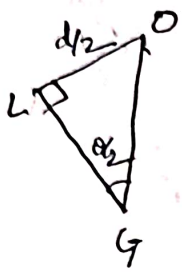
$$d_0 = D_1 \pm (R_2 - R_1)$$

dia of std. Cyl. Micrometer Reading over screw thread

⑤ effective dia. (Three wire method) screw thread micrometer



E — effective dia of screw thread
H — diameter over wire (Major)
Y — dia under wire (Min) $[H - 2d]$
d — dia of wire.



$$\triangle OLG \Rightarrow \sin(\theta/2) = \frac{OL}{OG}$$

$$d/2 \operatorname{cosec}(\theta/2) = OG$$

$$\triangle NIK \quad \tan \theta/2 = \frac{NI}{IK}$$

$$\cot \theta/2 = \frac{NI}{IK} \Rightarrow$$

$$\begin{aligned} NI &= M \\ IK &= P/2 \end{aligned}$$

$$\cot \theta/2 = M/P/2$$

$$\Rightarrow M = P/2 \cot \theta/2$$

$$\therefore RG = \frac{1}{2} \times M = \frac{1}{2} \times P/2 \cot \theta/2$$

$$\boxed{RG = P/4 \cot \theta/2}$$

$$OR + RG = OG \Rightarrow OR = OG - RG$$

$$= d/2 \operatorname{cosec} \theta/2 - P/4 \cot \theta/2$$

$$H = E + 2(OR) + 2(OG)$$

$$= E + 2 \left[d/2 \operatorname{cosec} \theta/2 - P/4 \cot \theta/2 \right] + 2(d/2)$$

$$\therefore H = E + d(\operatorname{cosec} \theta/2 + 1) - P/2 \cot \theta/2$$

(i) for Metric thread

$$\text{depth of thread} = 0.6495, \quad \theta = 60^\circ \quad \text{std. Nomencl}$$

$$\therefore \text{effective dia} = E = \underset{\text{Major Dia}}{D} - 0.6495 P$$

$$\therefore H = E + d(\operatorname{cosec} \theta/2 + 1) - P/2 \cot \theta/2$$

$$= D - 0.6495P + d(\operatorname{cosec} \theta/2 + 1) - P/2 \cot \theta/2 \quad \theta = 60^\circ$$

$$\therefore \boxed{H = D + 3d - 1.5155P}$$

(ii) Whitworth thread

$$\text{depth of thread} = 0.64 \quad \theta = 55^\circ$$

$$E = D - 0.64P$$

$$\therefore \boxed{H = D + 3.1656d - 1.6P}$$

Gear Parameters Measurement using Gear Tooth Vernier Caliper

① Outside dia — MSR + VSR + LC
(OD) $60 + 0 \times 0.02 = 60 \text{ mm}$

② Gear tooth thickness — MSR $4 + 10 \times 0.02 = 4.20 \text{ mm}$

③ Depth of gear tooth — $3 + 35 \times 0.02 = 3.70 \text{ mm}$

④ PCD (Pitch Circle dia)

Imaginary Circle — where tooth touches to another gear's teeth

$$D = T \times OD / T + 2$$

$$T = \text{No. of teeth} = 28$$

$$= 28 \times 60 / 28 + 2 = 56 \text{ mm.}$$

⑤ Module (M) — length of PCD per tooth

$$m = \frac{D}{T} = \frac{56}{28} = 2 \text{ mm}$$

⑥ Circular pitch: Arc distance measured around the pitch circle from flank of one tooth to the flank of next tooth [same side of next tooth].

$$CP = \frac{\pi D}{T} = \pi m = 3.14 \times 2 = 6.28 \text{ mm}$$

⑦ Addendum: Radial distance, measured from top land of tooth to the pitch circle.

$$\text{Add.} = 1m = 2 \text{ mm}$$

⑧ Clearance: Radial distance from tip of tooth to the bottom of mating tooth space when teeth are symmetrically engaged.

$$\text{std. values} = 0.157m \quad \text{or} \quad 0.25m$$

$$= 0.314m$$

$$0.50 \text{ mm}$$

⑨ Dedendum: Radial distance, measured from PCD to the bottom land of gear tooth

$$\text{Ded.} = \text{Add.} + \text{Clearance}$$

$$= 2 + 0.314 = 2.314 \text{ mm}$$

or

$$= 2 + 0.50 = 2.50 \text{ mm.}$$

∞ Chordal addendum — Set on vertical scale with help of slip gauges

$$d = \left(\frac{Tm}{2}\right) \left[1 + \left(\frac{2}{T}\right) - \cos\left(\frac{90}{T}\right)\right]$$

$$= \left(\frac{28 \times 2}{2}\right) \left[1 + \left(\frac{2}{28}\right) - \cos\left(\frac{90}{28}\right)\right]$$

$$= 2.044 \text{ mm by slip gauges.}$$

$$\frac{1.004}{1.04} \checkmark$$

Select tooth

Chordal thickness

$$W = MSR + VSR \times LC$$

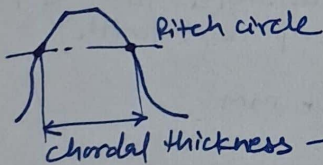
$$= 3 + 2 \times 0.02 = 3.04 \text{ mm}$$

3 Readings, then average.

Theoretical formula for chordal thickness

$$W = Tm \left[\sin\left(\frac{90}{T}\right)\right]$$

$$= 28 \times 2 \left[\sin\left(\frac{90}{28}\right)\right]$$



→ Tooth thickness measured along a chord passing through the points where the pitch circle crosses the tooth profile.

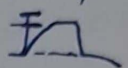
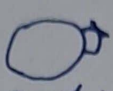
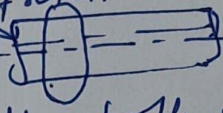
⑪ Runout : eccentricity in the Pitch circle

↳ give periodic vibration during each revolution of the gear

↳ give the tooth failures in gears.

Runout is measured by means of eccentricity testers.

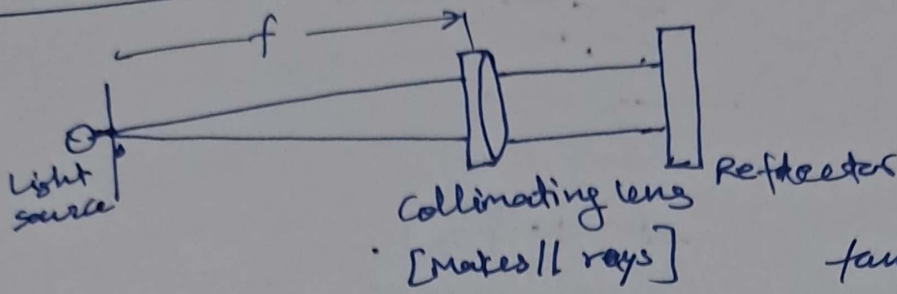
* Errors in Gear Measurement

1. Profile error: Max^m distance of any point on the tooth profile from and normal to the design profile. 
2. Cyclic error: it occurs during each revolution of the gear.  breakage of tooth.
3. Periodic error: error which occurs at regular intervals. 
4. Runout: total range of reading of a fixed indicator at a point to a rotating surface.
 - Radial runout: measured along $\perp$ to the axis of rotation
 - Axial ": " " " "
5. Eccentricity: diff. b/w axis of rotation & geometrical axis  gear rotates some out of axis.
6. Backlash: b/w mating tooth profile.
= tooth space — tooth thickness

Parkinson Gear Tester

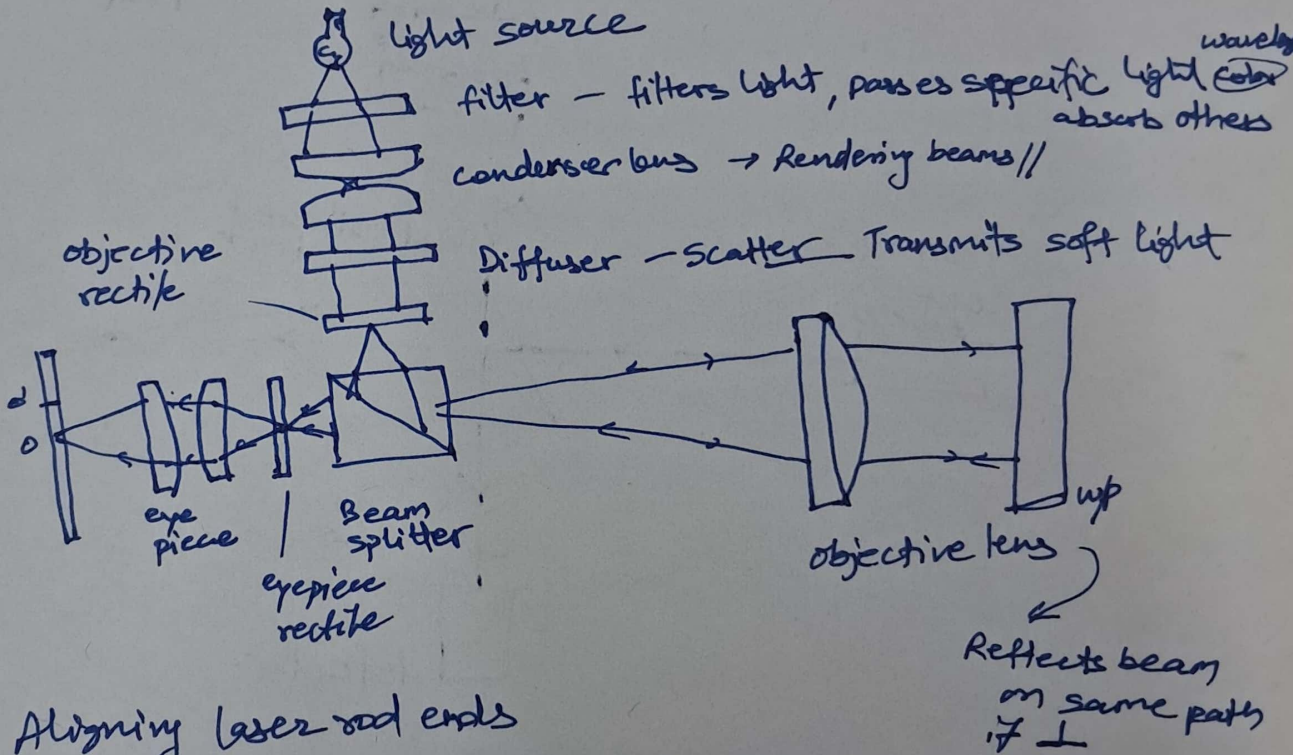
Auto collimator

Measuring
— Angle, straightness, flatness, squareness
+ parallelism.



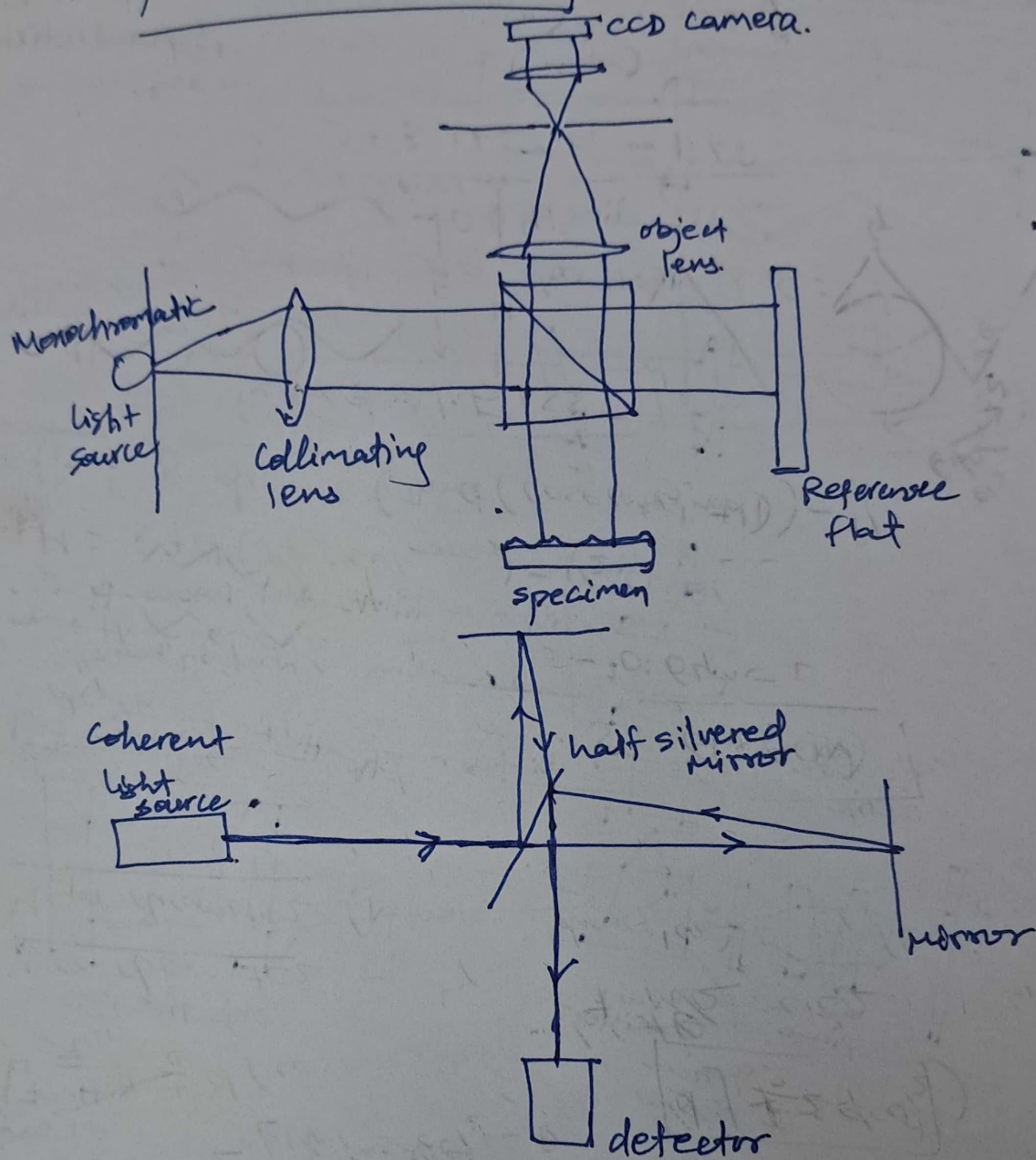
$$\text{magn} = \frac{x}{f} \Rightarrow 20 \times f = x$$

$$x = 20 \times f$$



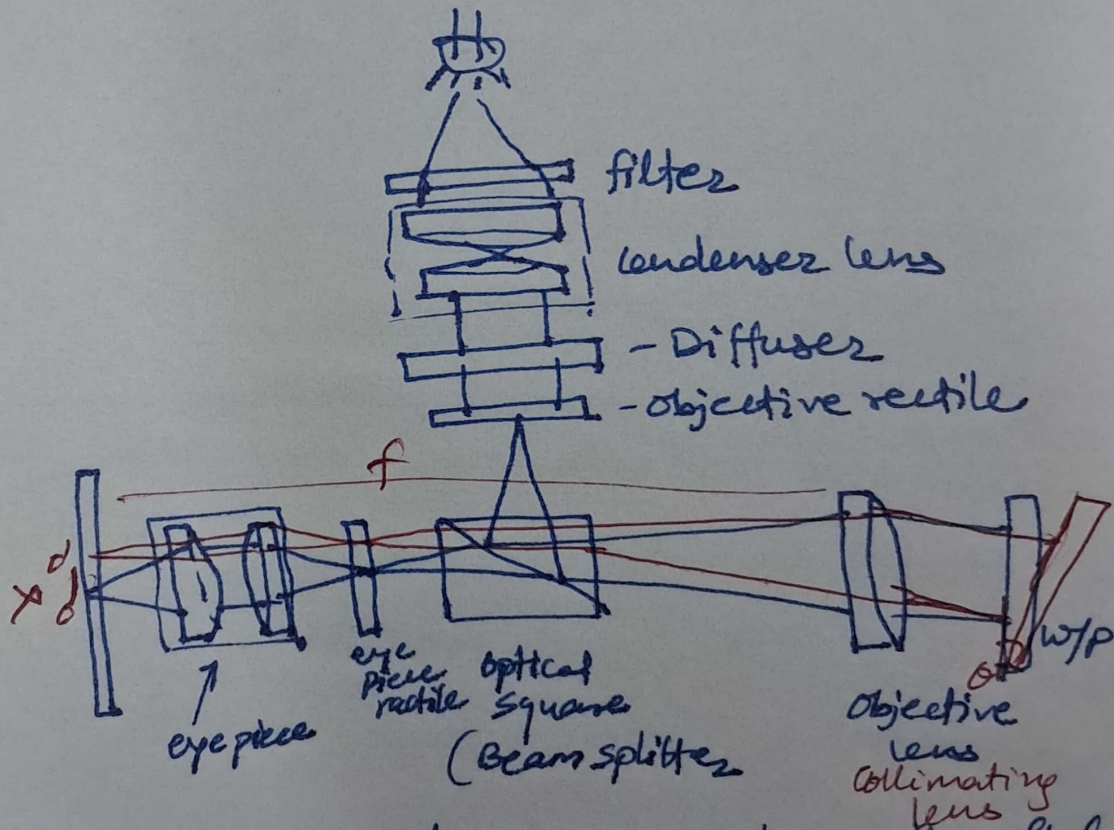
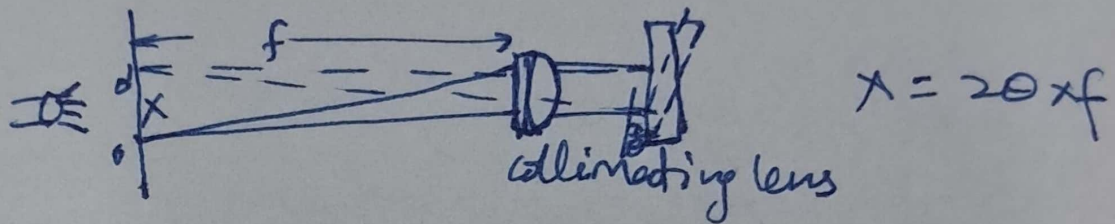
- Applⁿ —
- Aligning laser rod ends
 - checking face parallelism of optical windows + wedges
 - Monitoring angular movement over long periods of time
 - checking angular position in mech-system

Optical Interferometry — study surface structure



Autocollimator

- Visual — 4.85 micro radians
- Digital — 100 times more resolution
- Measure small angular diff.
- Used to align optical components
- Used to measure optical & mech. deflection



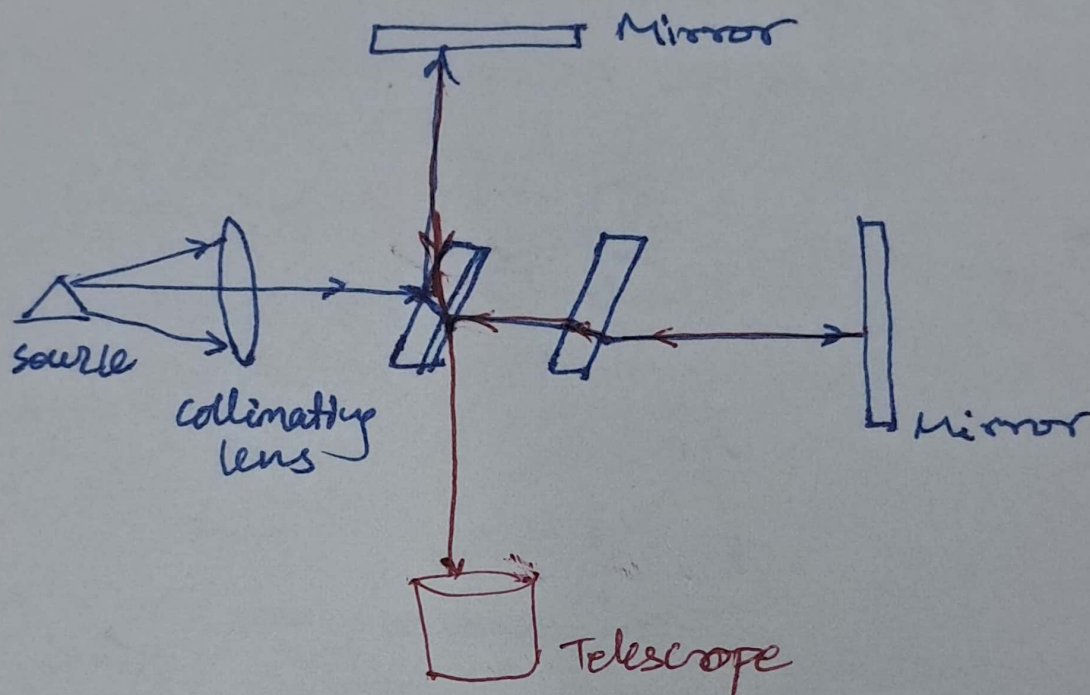
Diffuser — Scatter light rays to transmit soft lights

Appn

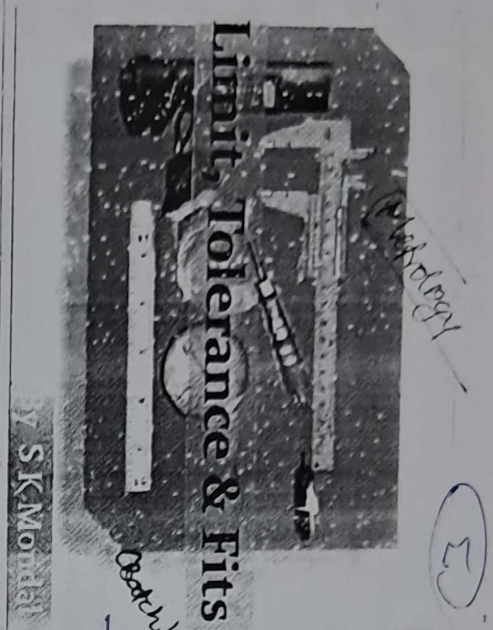
- Checking straightness of machine tool sideways
- Checking parallelism
- Checking squareness of column to base
- checking small linear displacements
- Measuring very small angles.

Optical Interferometer

- Used to study fine structure of spectral lines
- To determine wavelength of Monochromatic light



- Used to make very small measurements that are not achievable any other way
- length, surface irregularities



Terminology

Contd....

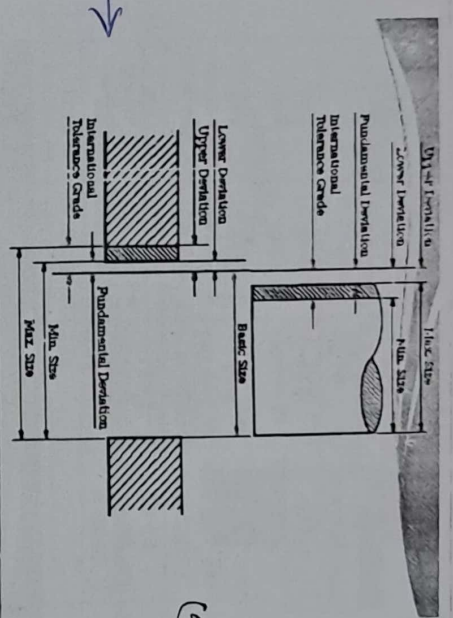
- **Limits of sizes:** There are two extreme permissible sizes for a dimension of the part. The largest permissible size for a dimension is called upper or high or maximum limit, whereas the smallest size is known as lower or minimum limit.
- **Tolerance**

$$T = U.L - L.L$$
 - The difference between the upper limit and lower limit.
 - It is the maximum permissible variation in a dimension.
 - The tolerance may be unilateral or bilateral.

Terminology

Contd....

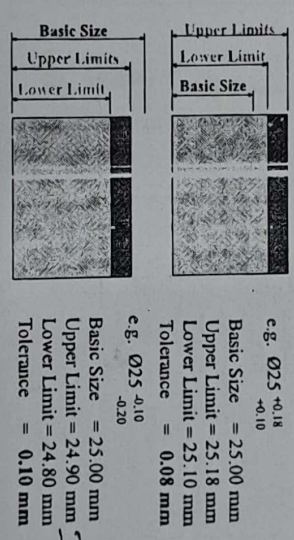
- **Zero line:** A straight line corresponding to the basic size. The deviations are measured from this line.
- **Deviation:** Is the algebraic difference between a size (actual, max. etc.) and the corresponding basic size.
- **Actual deviation:** Is the algebraic difference between an actual size and the corresponding basic size.
- **Upper deviation:** Is the algebraic difference between the maximum size and the basic size.



Terminology

Contd....

Unilateral limits occurs when both maximum limit and minimum limit are either above or below the basic size.



e.g. $\varnothing 25^{+0.18}_{-0.10}$
 Basic Size = 25.00 mm
 Upper Limit = 25.18 mm
 Lower Limit = 25.10 mm
 Tolerance = 0.08 mm

Terminology

Contd....

- **Lower deviation:** Is the algebraic difference between the minimum size and the basic size.
- **Mean deviation:** Is the arithmetical mean of upper and lower deviations.
- **Fundamental deviation:** This is the deviation, either the upper or the lower deviation, which is nearest one to zero line for either a hole or shaft.

Terminology

- **Nominal size:** Size of a part specified in the drawing.
- **Basic size:** Size of a part to which all limits of variation (i.e. tolerances) are applied.

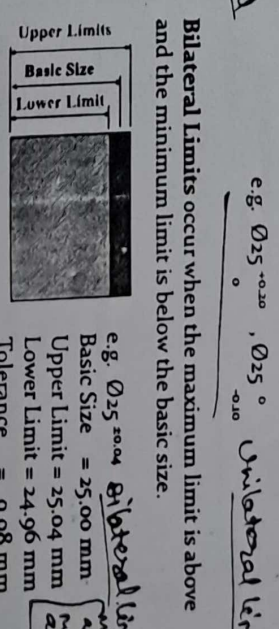
Actual size: Actual measured dimension of the part.

The difference between the basic size and the actual size should not exceed a certain limit, otherwise it will interfere with the interchangeability of the mating parts.

Terminology

Contd....

For Unilateral limits, a case may occur when one of the limits coincides with the basic size.



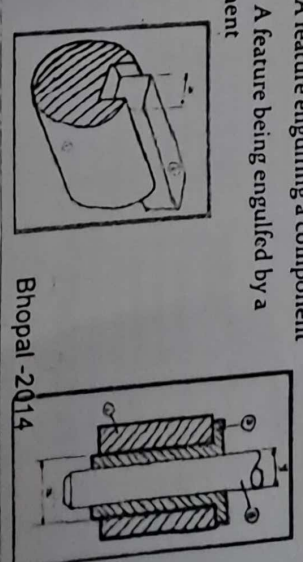
e.g. $\varnothing 25^{+0.20}_{0}$, $\varnothing 25^{0}_{-0.10}$
 Basic Size = 25.00 mm
 Upper Limit = 25.04 mm
 Lower Limit = 24.96 mm
 Tolerance = 0.08 mm

Fit

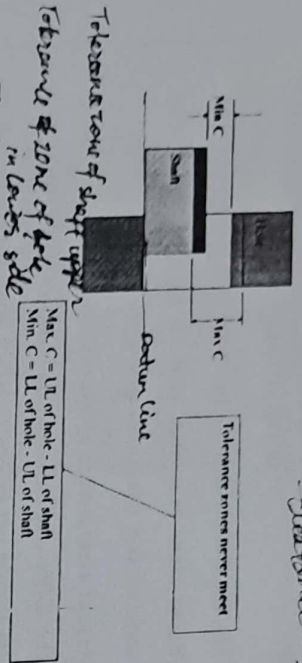
Fits: (assembly condition between "Hole" & "Shaft")

Hole - A feature engulfing a component

Shaft - A feature being engulfed by a component

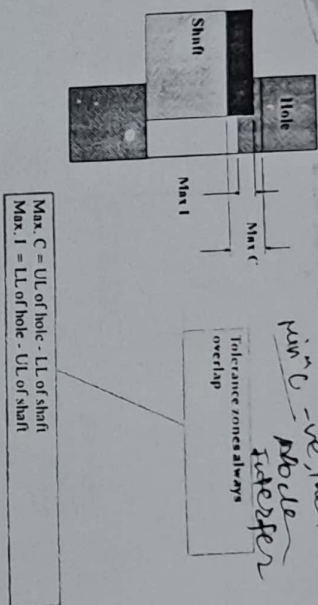


Clearance Fits



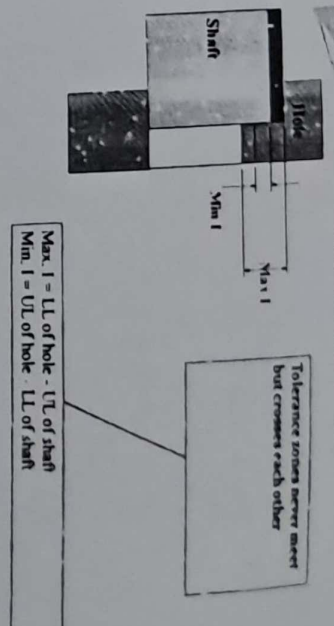
The clearance fits may be slide fit, easy sliding fit, running fit, slack running fit and loose running fit.

Transition Fits



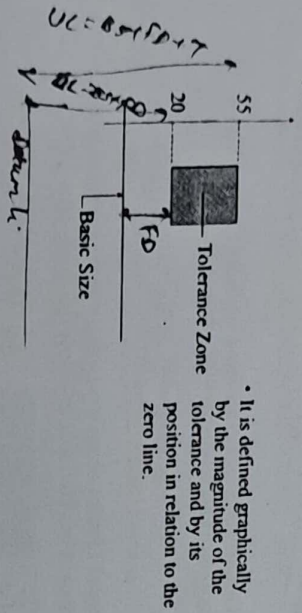
The transition fits may be push fit, tight fit and push fit.

Interference Fits



The interference fits may be shrink fit, heavy drive fit and light drive fit.

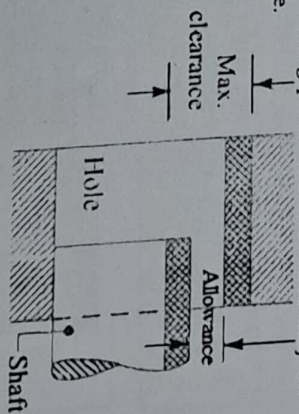
Tolerance Zone



It is defined graphically by the magnitude of the tolerance and by its position in relation to the zero line.

Allowance

It is minimum clearance or maximum interference. It is the intentional difference between the basic dimensions of the mating parts. The allowance may be positive or negative.



Limits and Fits

Limits and fits comprises 18 grades of fundamental tolerances for both shaft and hole, designated as IT01, IT0 and IT1 to IT16. These are called standard tolerances. (IS-919) But ISO 286 specify 20 grades upto IT18

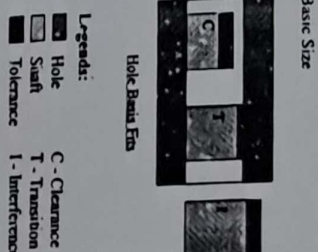
There are 25 (IS 919) and 28 (ISO 286) types of fundamental deviations.

Hole: A, B, C, CD, D, E, EF, F, FG, G, H, J, JS, K, M, N, P, R, S, T, U, V, X, Y, Z, ZA, ZB, ZC.

Shaft: a, b, c, cd, d, e, ef, f, fg, g, h, js, k, m, n, p, r, s, t, u, v, x, y, z, za, zb, zc.

A unilateral hole basis system is recommended but if necessary a unilateral or bilateral shaft basis system may also be used

5. BASIS OF FITS - HOLE BASIS SYSTEM



In this system, the basic diameter of the hole is constant while the shaft size varies according to the type of fit.

Tolerance Designation (ISO)

Tolerance on a shaft or a hole can also be calculated by using the formulas provided by ISO.

$$T = K \times i$$

where,

T is the tolerance (in μm)

$i = 0.45\sqrt[3]{D} + 0.001D$ (unit tolerance, in μm)

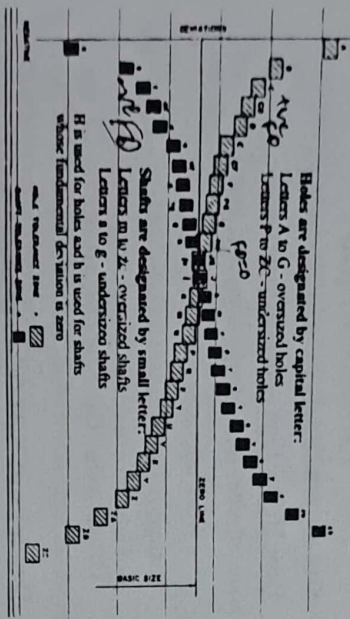
$D = \sqrt[3]{D_1 D_2}$ (D_1 and D_2 are the nominal sizes marking the beginning and the end of a range of sizes, in mm)

$K = 10(1.6)^{(IT_n - IT_0)}$ [For IT6 to IT16]

Diameter Steps

Above (mm)	Up to and including (mm)
3	3
5	6
6	10
10	18
18	30
30	50
50	80
80	120
120	180
180	250
250	315
315	400
400	500

is chosen to locate the tolerance zone w.r.t. the zero line

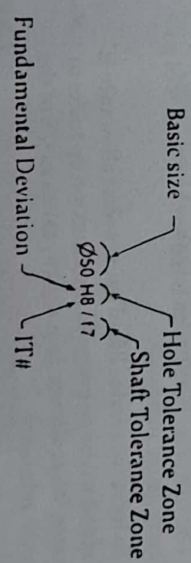
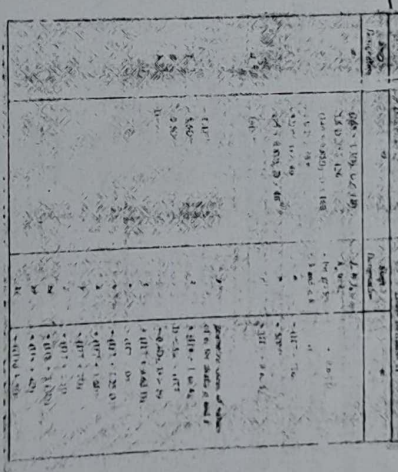


- For hole, H stands for a dimension whose lower deviation refers to the basic size. The hole H for which the lower deviation is zero is called a basic hole.
- Similarly, for shafts, h stands for a dimension whose upper deviation refers to the basic size. The shaft h for which the upper deviation is zero is called a basic shaft.
- A fit is designated by its basic size followed by symbols representing the limits of each of its two components, the hole being quoted first.
- For example, 100 H6/g5 means basic size is 100 mm and the tolerance grade for the hole is 6 and for the shaft is 5.

For IES, GATE, PSUs

Value of the Tolerance	
IT01	IT0
0.3 + 0.008D	0.5 + 0.012D
IT1	IT2
0.8 + 0.02D	$r = 10^{1/8}$
IT3	IT4
ar^2	ar^3
IT5	IT6
$ar^4 = 71$	$ar^5 = 101$
IT7	IT8
$ar^6 = 161$	$ar^7 = 251$
IT9	IT10
$ar^8 = 401$	$ar^9 = 641$
IT11	IT12
$ar^{10} = 1001$	$ar^{11} = 4001$
IT13	IT14
$ar^{12} = 2501$	$ar^{13} = 6401$
IT15	IT16
$ar^{14} = 10001$	$ar^{15} = 64001$

Fundamental Deviations



Grades of Tolerance

- It is an indication of the level of accuracy.
- IT01 to IT4 - For production of gauges, plug gauges, measuring instruments
- IT5 to IT7 - For fits in precision engineering applications
- IT8 to IT11 - For General Engineering
- IT12 to IT14 - For Sheet metal working or press working
- IT15 to IT16 - For processes like casting, general cutting work

Calculation for Upper and Lower Deviation

- For Shaft
 - $ei = es - IT$ lower deviation
 - $es = ei + IT$
- For Hole
 - $EI = ES - IT$
 - $ES = EI + IT$
- es = upper deviation of shaft
- ei = lower deviation of shaft
- ES = upper deviation of hole
- EI = lower deviation of hole

Recommended Selection of Fits

TYPE OF FIT	SLUOT AND TOLERANCE	HOLE AND TOLERANCE			
		H7	H8	H9	H11
CLEARANCE	F 11				
	F 10				
	F 9				
TRANSITION	F 8				
	F 7				
	F 6				
INTERFERENCE	F 5				
	F 4				
	F 3				

Selective Assembly

- All the parts (hole & shaft) produced are measured and graded into a range of dimensions within the tolerance groups.
- Reduces the cost of production

Handwritten notes:
1.5 to 2.5 mm
10 to 15 mm
For more cost of production

Limit Gauges

- Plug gauge:** used to check the holes. The GO plug gauge is the size of the low limit of the hole while the NOT GO plug gauge corresponds to the high limit of the hole.
- Snap, Gap or Ring gauge:** used for gauging the shaft and male components. The Go snap gauge is of a size corresponding to the high (maximum) limit of the shaft, while the NOT GO gauge corresponds to the low (minimum limit).

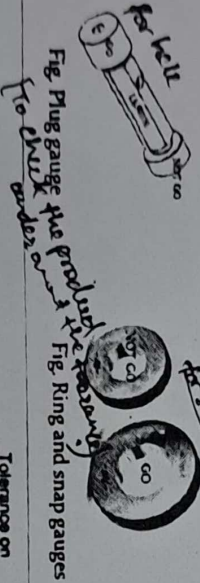
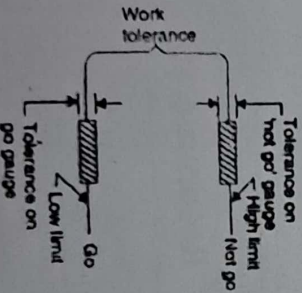


Fig. Ring and snap gauges

- Bilateral system:** in this system, the GO and NO GO gauge tolerance zones are bisected by the high and low limits of the work tolerance zone.



Taking example as above:

$$\therefore \text{Dimension of 'GO' Plug gauge} = 24.98 \pm 0.002 \text{ mm}$$

$$\text{Dimension of 'NOT GO' Plug gauge} = 25.02 \pm 0.002 \text{ mm}$$

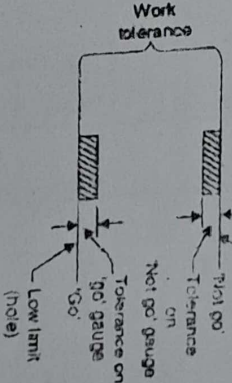
For IES, GATE, PSUs

Interchangeability

- Term employed for the mass production of identical items within the prescribed limits of sizes.
- If the variation of items are within certain limits, all parts of equivalent size will be equally fit for operating in machines and mechanisms and the mating parts will give the required fitting.
- This facilitates to select at random from a large number of parts for an assembly and results in a considerable saving in the cost of production.

Allocation of manufacturing tolerances

- Unilateral system:** gauge tolerance zone lies entirely within the work tolerance zone.
- work tolerance zone becomes smaller by the sum of the gauge tolerance.



- Wear allowance:** GO gauges which constantly rub against the surface of the parts in the inspection are subjected to wear and lose their initial size.
- The size of go plug gauge is reduced while that of go snap gauge increases.
- To increase service life of gauges wear allowance is added to the go gauge in the direction opposite to wear. Wear allowance is usually taken as 1% of the work tolerance.
- Wear allowance is applied to a nominal diameter before gauge tolerance is applied.

Tolerance Sink

- A design engineer keeps one section of the part blank (without tolerance) so that production engineer can dump all the tolerances on that section which becomes most inaccurate dimension of the part.
- Position of sink can be changing the reference point.
- Tolerance for the sink is the cumulative sum of all the tolerances and only like minded tolerances can be added i.e. either equally bilateral or equally unilateral.

Example

Size of the hole to be checked $25 \pm 0.02 \text{ mm}$
Here, Higher limit of hole = 25.02 mm
Lower limit of hole = 24.98 mm

$$\text{Work tolerance} = 0.04 \text{ mm}$$

$$\therefore \text{Gauge tolerance} = 10\% \text{ of work tolerance} = 0.004 \text{ mm}$$

$$\therefore \text{Dimension of 'GO' Plug gauge} = 24.98 \pm 0.004 \text{ mm}$$

$$\therefore \text{Dimension of 'NOT GO' Plug gauge} = 25.02 \pm 0.004 \text{ mm}$$

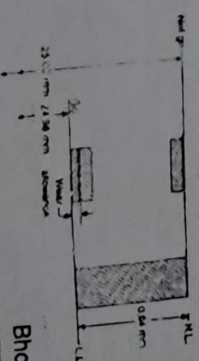
- Taking example of above:

$$\therefore \text{Wear Allowance} = 5\% \text{ of work tolerance} = 0.002 \text{ mm}$$

$$\therefore \text{Nominal size of GO plug gauge} = 24.98 + 0.002 \text{ mm}$$

$$\therefore \text{Dimension of 'GO' Plug gauge} = 24.982 \pm 0.004 \text{ mm}$$

$$\therefore \text{Dimension of 'NOT GO' Plug gauge} = 25.02 \pm 0.004 \text{ mm}$$



Taylor's Principle

This principle states that the GO gauge should always be so designed that it will cover the maximum metal condition (MMC) of as many dimensions as possible in the same limit gauges, whereas a NOT GO gauges to cover the minimum metal condition of one dimension only.

Vernier Caliper

- A vernier scale is an auxiliary scale that slides along the main scale.
- The vernier scale is that a certain number n of divisions on the vernier scale is equal in length to a different number (usually one less) of main-scale divisions.

$$nV = (n-1)S$$

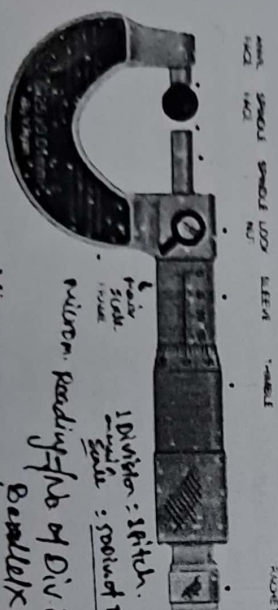
where n = number of divisions on the vernier scale

V = The length of one division on the vernier scale

and S = Length of the smallest main-scale division

- Least count is applied to the smallest value that can be read directly by use of a vernier scale.

$$\text{Least count} = S - V = \frac{1}{n} S$$

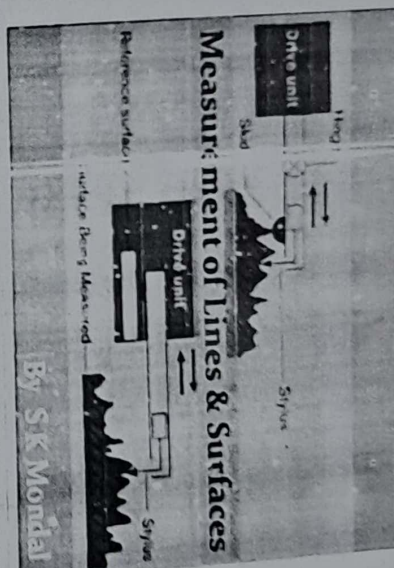


Micrometer

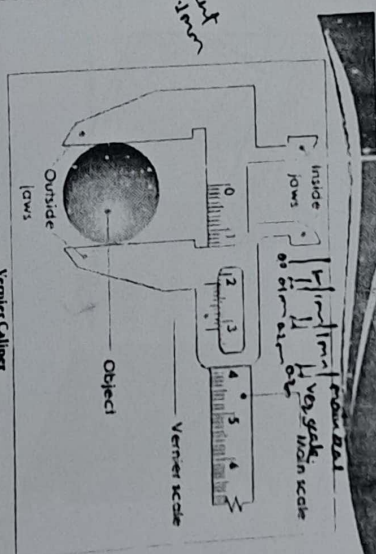
Reading of Div in the
Bore Gauge Pitch
+ Number Reading of Div
No of Divs in the Thimble

For IES, GATE, PSUs

Measurement of Lines & Surfaces



By S.K. Mondal



Vernier Caliper

Bore Gauge: used for measuring bores of different sizes ranging from small-to-large sizes.

- Provided with various extension arms that can be added for different sizes.

Linear measurements

Some of the instruments used for the linear measurements are:

- Rules (Steel)
- Vernier
- Micrometer - In worst step
- Height gauge
- Bore gauge
- Dial indicator
- Slip gauges or gauge blocks

Metric Micrometer

- A micrometer allows a measurement of the size of a body. It is one of the most accurate mechanical devices in common use.
- It consists a main scale and a thimble

Method of Measurement

Step-I: Find the whole number of mm in the barrel

Step-II: Find the reading of barrel and multiply by 0.01

Step-III: Add the value in Step-I and Step-II

- Dial indicator:** Converts a linear displacement into a radial movement to measure over a small range of movement for the plunger.
- The typical least count that can be obtained with suitable gearing dial indicators is 0.01 mm to 0.001 mm.
- It is possible to use the dial indicator as a comparator by mounting it on a stand at any suitable height.

Bhopal 2014

- Applications of dial indicator include:**
- centering workpieces to machine tool spindles
 - offsetting lathe tail stocks
 - aligning a vice on a milling machine
 - checking dimensions

Slip Gauges or Gauge blocks

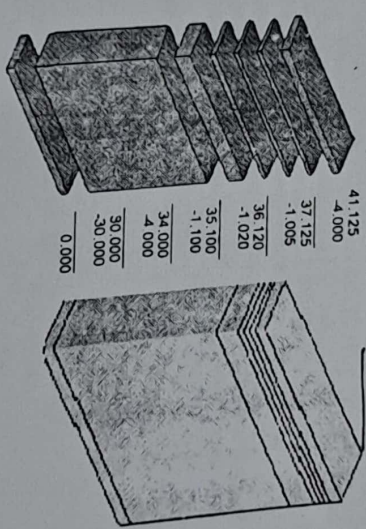
- These are small blocks of alloy steel.
- Used in the manufacturing shops as length standards.
- Not to be used for regular and continuous measurement.
- Rectangular blocks with thickness representing the dimension of the block. The cross-section of the block is usually 32 mm x 9 mm.
- Are hardened and finished to size. The measuring surfaces of the gauge blocks are finished to a very high degree of finish, flatness and accuracy.

- Come in sets with different number of pieces in a given set to suit the requirements of measurements.
- A typical set consisting of 88 pieces for metric units is shown in.
- To build any given dimension, it is necessary to identify a set of blocks, which are to be put together.
- Number of blocks used should always be the smallest.
- Generally the top and bottom Slip Gauges in the pile are 2 mm wear gauges. This is so that they will be the only ones that will wear down, and it is much cheaper to replace two gauges than a whole set.

To make up a Slip Gauge pile to 41.125 mm

- A Slip Gauge pile is set up with the use of simple maths.
- Decide what height you want to set up, in this case 41.125mm.
- Take away the thickness of the two wear gauges, and then use the gauges in the set to remove each place of decimal in turn, starting with the lowest

To make up a Slip Gauge pile to 41.125 mm



Comparators

- Comparator is another form of linear measuring method, which is quick and more convenient for checking large number of identical dimensions.
- During the measurement, a comparator is able to give the deviation of the dimension from the set dimension.
- Cannot measure absolute dimension but can only compare two dimensions.
- Highly reliable.
- To magnify the deviation, a number of principles are used such as mechanical, optical, pneumatic and electrical.

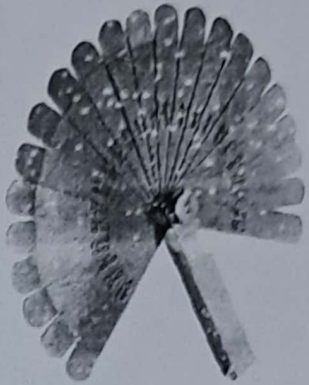
A Metric slip gauge set (88 Pieces)

Slip gauges size or range, mm	Increment, mm	Number of Pieces
1.005	-	1
1.001 to 1.009	0.001	9
1.010 to 1.490	0.010	49
0.500 to 9.500	0.500	19
10 to 100	10.000	10



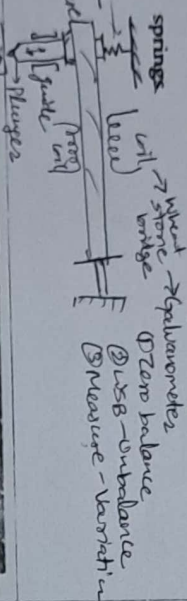
Fig. Principle of comparator

Feeler Gauge



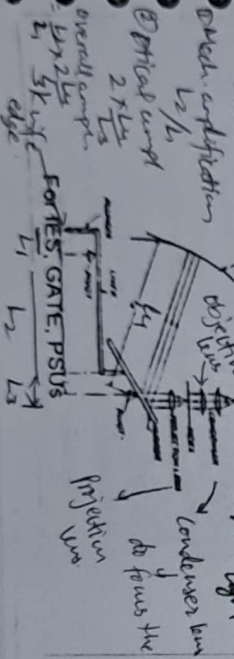
Sigma Mechanical Comparator

The Sigma Mechanical Comparator uses a partially wrapped band wrapped about a driving drum to turn a pointer needle. The assembly provides a frictionless movement with a resistant pressure provided by the springs.



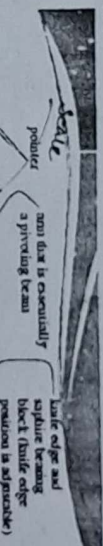
Optical Comparators

These devices use a plunger to rotate a mirror. A light beam is reflected off that mirror, and simply by the virtue of distance, the small rotation of the mirror can be converted to a significant translation with little friction.



Limit Gauges

Gauge	For Measuring
Snap Gauge	External Dimensions
Plug Gauge	Internal Dimensions
Taper Plug Gauge	Taper hole
Ring Gauge	External Diameter
Gap Gauge	Gaps and Grooves
Radius Gauge	Gauging radius
Thread pitch Gauge	External Thread

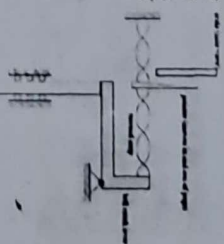


Pneumatic Comparators

- Flow type:
- The float height is essentially proportional to the air that escapes from the gauge head
- Master gauges are used to find calibration points on the scales
- The input pressure is regulated to allow magnification adjustment

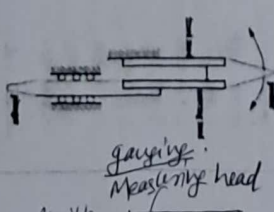
Mechanical Comparators

The Mikrokator principle greatly magnifies any deviation in size so that even small deviations produce large deflections of the pointer over the scale.

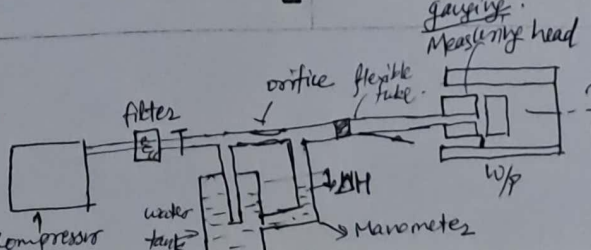
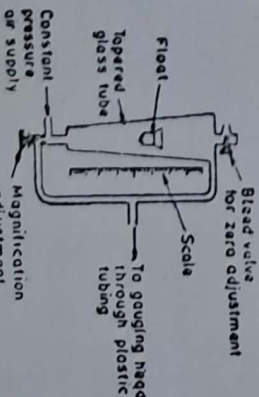


Mechanical Comparators

- The Eden-Rolt Reed system uses a pointer attached to the end of two reeds. One reed is pushed by a plunger, while the other is fixed. As one reed moves relative to the other, the pointer that they are commonly attached to will deflect.



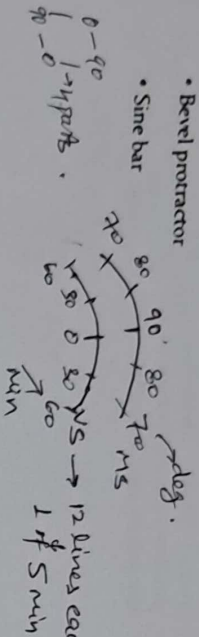
Pneumatic Comparators



Angular Measurement

This involves the measurement of angles of tapers and similar surfaces. The most common angular measuring tools are:

- Bevel protractor
- Sine bar



Sine Bar

- When a reference for a non-square angle is required, a sine bar can be used.
- Basically a sine bar is a bar of known length. When gauge blocks are placed under one end, the sine bar will tilt to a specific angle.
- Knowing the height differential of the two rollers in alignment with the workpiece, the angle can be calculated using the sine formula.
- A sine bar is specified by the distance between the centre of the two rollers, i.e. 100 mm, 200 mm, & 300 mm. the various part of the sine bar are hardened before grinding & lapping.



Bevel Protractor

- Is part of the machinist's combination square.
- The flat base of the protractor helps in setting it firmly on the workpiece and then by rotating the rule, it is possible to measure the angle. It will typically have a discrimination of one degree.

12 div. of VS = 23 div of MS

$$1 \text{ MSD} = 1^\circ \quad 1' = 60''$$

$$1 \text{ VSD} = 5' \quad 1' = 60''$$

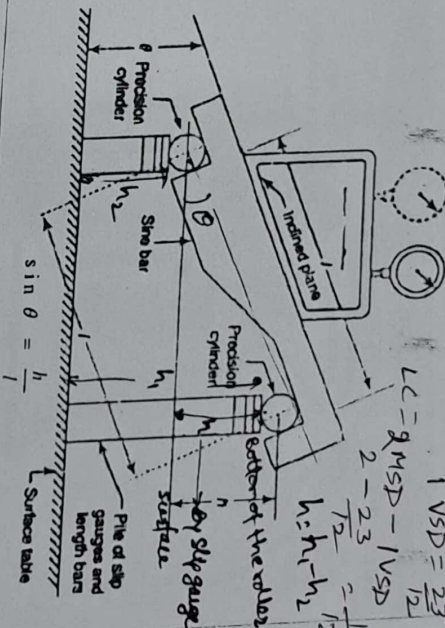
$$1 \text{ VSD} = \frac{23}{12} \text{ MSD}$$

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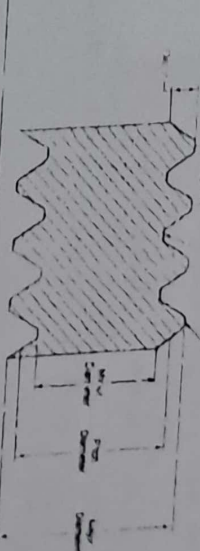
$$1 \text{ VSD} = \frac{23}{12} \text{ MSD}$$

$$1 \text{ VSD} = \frac{23}{12} \text{ MSD}$$



Three-Wire Method

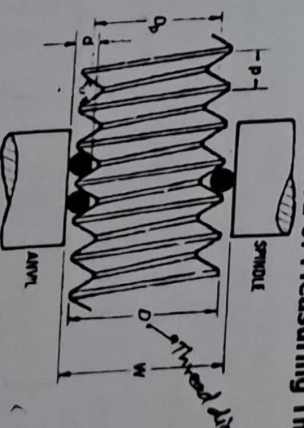
- Three wires of equal diameter placed in thread, two on one side and one on other side
- Standard micrometer used to measure distance over wires (M)
- Different sizes and pitches of threads require different sizes of wires



Thread Measurements

- Threads are normally specified by the major diameter.
- Though there are a large variety of threads used in engineering, the most common thread encountered is the metric V-thread shown in Fig.

The Three-Wire Method of Measuring Threads



D_p = pitch diameter or Effective diameter
 P = pitch of thread, and W = wire diameter

Distance W over the outer edge

$$W = D_r + d \left(1 + \cos \sec \frac{\alpha}{2} \right) - \frac{P}{2} \cot \frac{\alpha}{2}$$

For ISO metric thread, $\alpha = 60$ and $D_r = D - 0.6496p$

$$W = D + 3d - 1.5156p$$

Best wire size, μ INP

$$d = \frac{P}{2} \sec \frac{\alpha}{2}$$

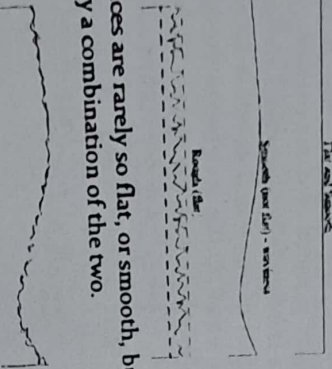
For ISO metric thread, $\alpha = 60$

$$d = 0.5774p$$

Effective dia
→ Pitch dia.

Surface geometry can be quantified a few different ways.

- Real surfaces are rarely so flat, or smooth, but most commonly a combination of the two.



- **Waviness:** refers to those surface irregularities that have a greater spacing than that of roughness width.
- Determined by the height of the waviness and its width.
- The greater the width, the smoother is the surface and thus is more desirable.
- **Lay direction:** is the direction of the predominant surface pattern produced on the workpiece by the tool marks.
- **Flaw:** are surface irregularities that are present which are random and therefore will not be considered.

For IES, GATE, PSUs

Measurement of Surfaces

→ Roughness
→ Profile (Roughness + waviness)

- **Roughness height:** is the parameter with which generally the surface finish is indicated. It is specified either as arithmetic average value or the root mean square value.
- **Roughness width:** is the distance parallel to the nominal part surface within which the peaks and valleys, which constitutes the predominant pattern of the roughness.
- **Roughness width cut-off:** is the maximum width of the surface that is included in the calculation of the roughness height.

Surfaces

- No surface is perfectly smooth, but the better the surface quality, the longer a product generally lasts, and the better it performs.
- Surface texture can be difficult to analyse quantitatively.
- Two surfaces may be entirely different, yet still provide the same CLA (Ra) value.

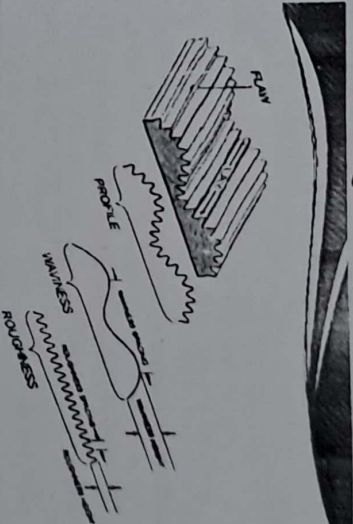
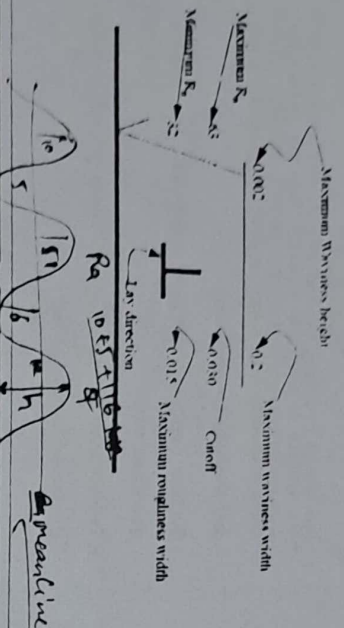


Diagram	Symbol	Description
		Parallel lay: Lay parallel to the Surface. Surface is produced by shaping, planing etc.
		Perpendicular lay: Lay perpendicular to the Surface. Surface is produced by shaping and planing
		Crossed lay: Lay angular in both directions. Surface is produced by grinding, honing.

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Diagram	Symbol	Description
		Multidirectional lay: Lay multidirectional. Surface is produced by grinding, lapping, super finishing.
		Circular lay: Approximately circular relative to the center. Surface is produced by facing.
		Radial lay: Approximately radial relative to the center of the nominal surface.

Representation of Surface Roughness



Evaluation of Surface Roughness

1. Centre line average (CLA) or arithmetic mean deviation denoted as R_a .
2. Root mean square value (R_q) : rms value.
3. Maximum peak to valley roughness (R_{max})
4. The average of the five highest peak and five deepest valleys in the sample.
5. The average or leveling depth of the profile.

Arithmetical Average:

- Measured for a specified area and the figures are added together and the total is then divided by the number of measurements taken to obtain the mean or arithmetical average (R_a).
- It is also sometimes called the centre line average or CLA value. This in equation form is given by

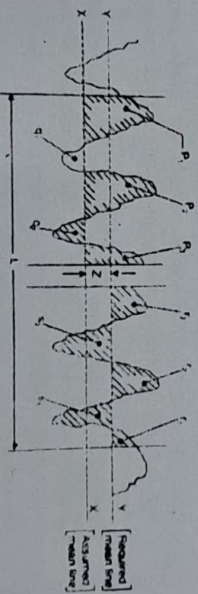
$$R_a = \frac{1}{L} \int_0^L |y(x)| dx \approx \frac{1}{N} \sum |y_i|$$

For IES, GATE, PSUs

Roughness	Roughness Grade	Ro	Finest Symbol
50 / 12.5 R	N12		▽
25 / 6.3 R	N11		▽
12.5 / 3.2 R	N10		▽
6.3 / 1.6 R	N9		▽
3.2 / 0.8 R	N8		▽
1.6 / 0.4 R	N7		▽
0.8 / 0.2 R	N6		▽
0.4 / 0.1 R	N5		▽
0.2 / 0.05 R	N4		▽
0.1 / 0.025 R	N3		▽
0.05 / 0.0125 R	N2		▽
0.025 / 0.0063 R	N1		▽

Determination of Mean Line

- **M-System:** After plotting the characteristic of any surface a horizontal line is drawn by joining two points. This line is shifted up and down in such a way that 50% area is above the line and 50% area is below the line



- The other parameter that is used sometimes is the root mean square value of the deviation in place of the arithmetic average, This in expression form is

$$R_{RMS} = \sqrt{\frac{1}{N} \sum y_i^2}$$

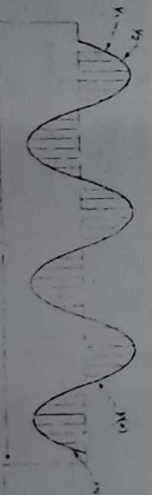
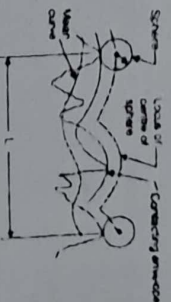


Fig. Surface roughness parameters

- **Waviness height** - the distance from a peak to a valley.
- **Waviness width** - the distance between peaks or valleys
- **Roughness width cutoff** - a value greater than the maximum roughness width that is the largest separation of surface irregularities included in the measurements. Typical values are (0.003, 0.010, 0.030, 0.100, 0.300)
- **Lay** - the direction the roughness pattern should follow
- Stylus travel is perpendicular to the lay specified.

Determination of Mean Line

- **E-System: (Envelope System)** A sphere of 25 mm diameter is rolled over the surface and the locus of its centre is being traced out called envelope. This envelope is shifted in downward direction till the area above the line is equal to the area below the line. This is called mean envelope and the system of datum is called E-system.



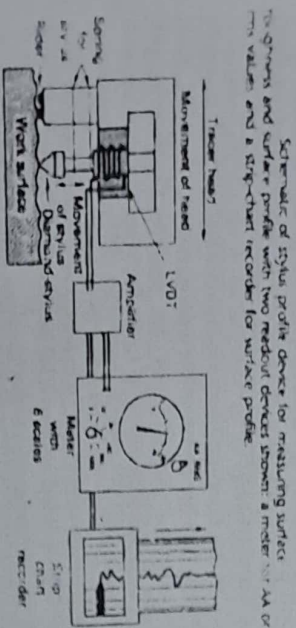
Methods of measuring Surface Roughness

- There are a number of useful techniques for measuring surface roughness:
- **Observation and touch** - the human finger is very perceptive to surface roughness
 - **Stylus based equipment** - very common
 - **Interferometry** - uses light wave interference patterns (discussed later)

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Observation Methods

- Human perception is highly relative.
- To give the human tester a reference for what they are touching, commercial sets of standards are available.
- Comparison should be made against matched identical processes.
- One method of note is the finger nail assessment of roughness and touch method.



Non-contact Profilometers

- An optical profilometer is a non-contact method for providing much of the same information as a stylus based profilometer.
- There are many different techniques which are currently being employed, such as laser triangulation (triangulation sensor), confocal microscopy and digital holography.

Stylus Equipment

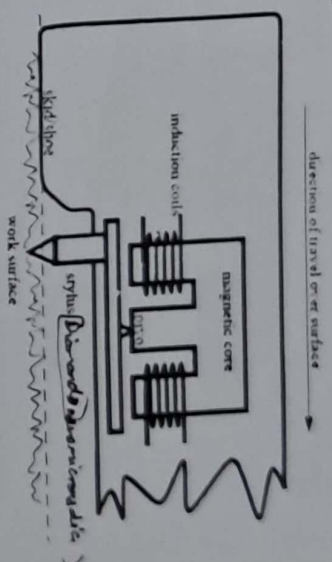
- uses a stylus that tracks small changes in surface height, and a skid that follows large changes in surface height.
- The relative motion between the skid and the stylus is measured with a magnetic circuit and induction coils.
- One example of this is the Brown & Sharpe Surfcom unit.

Profilometer

- measuring instrument used to measure a surface's profile, in order to quantify its roughness.
- Vertical resolution is usually in the nanometre level, though lateral resolution is usually poorer.

Advantages of optical Profilometers

- Because the non-contact profilometer does not touch the surface the scan speeds are dictated by the light reflected from the surface and the speed of the acquisition electronics.
- Optical profilometers do not touch the surface and therefore cannot be damaged by surface wear or careless operators.



Contact profilometers

- A diamond stylus is moved vertically in contact with a sample and then moved laterally across the sample for a specified distance and specified contact force.
- A profilometer can measure small surface variations in vertical stylus displacement as a function of position.
- The radius of diamond stylus ranges from 20 nanometres to 25 μm .

Optical Flats

- Optical-grade glass structures or clear fused quartz lapped and polished to be extremely flat on one or both sides.
- Used with a monochromatic light to determine the flatness of other optical surfaces by interference.
- When a flat surface of another optic is placed on the optical flat, interference fringes are seen due to interference in the tiny gap between the two surfaces.
- The spacing between the fringes is smaller where the gap is changing more rapidly, indicating a departure from flatness in one of the two surfaces, in a similar way to the contour lines on a map.

Diagram illustrating the experimental setup for measuring the moment of inertia of a rectangular plate. A horizontal beam is supported by a central vertical axis. A rectangular plate is suspended from the left end of the beam, and a standard weight is suspended from the right end. The distance from the axis to the center of the plate is labeled l . The plate is labeled "Steel plate" and the weight is labeled "Standard".

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- The distance of air gap between two successive fringes is given by $= \frac{\lambda}{2}$
- Distance of air gap of interference fringe of n^{th} order is $= \frac{n\lambda}{2}$

$$\Delta h = \frac{n\lambda l}{2} \quad \text{in cm}$$

n = number of fringes / cm
 Δh = The difference of height between gauges
 λ = wavelength of monochromatic light



Climometer

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Talysurf

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- Figure 1 consists of two schematic diagrams, (a) and (b), illustrating the interferometric measurement of a variable flat optoelectronic component.
- Diagram (a) shows a normally flat component. A light source (represented by a circle with a cross) emits light that passes through a lens and is reflected by the component. The reflected light is then collected by another lens and directed to a detector. The resulting interferogram shows concentric circular fringes, indicating a flat surface. Labels include "Normally flat component", "Optical flat", and "Interfering with reference beam".
- Diagram (b) shows a component with a step. The component has a flat top surface and a step of height 0.25 mm . The interferogram shows fringes at the step edge, indicating the presence of the step. Labels include "Optical flat", "Fringes at step edge", "Step height 0.25 mm ", and "Reference beam".

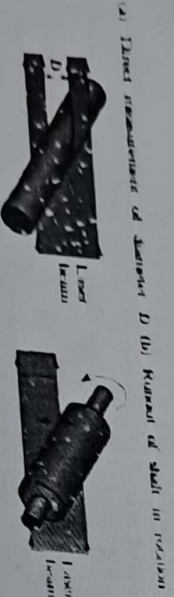


Autocollimator

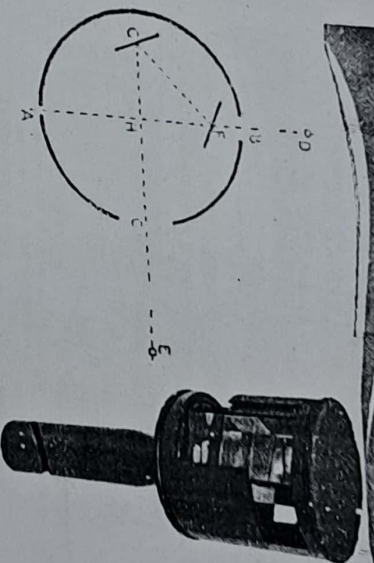
- An optical instrument for non-contact measurement of angles.
- Used to align components and measure deflections in optical or mechanical systems.
- An autocollimator works by projecting an image onto a target mirror, and measuring the deflection of the returned image against a scale, either visually or by means of an electronic detector.
- A visual autocollimator can measure angles as small as 0.5 arcsecond, while an electronic autocollimator can be up to 100 times more accurate.

Optical Square

- An Optical square consists of a small cylindrical metal box, about 5 cm in diameter and 12.5 cm deep, in which two mirrors are placed at an angle of 45° to each other and at right angles to the plane of the instrument.
- One mirror (horizon glass) is half silvered and other (index glass) is wholly silvered.
- The optical square belongs to a reflecting instruments which measure angles by reflection. Angle between the first incident ray and the last reflected ray is 90° .
- Used to find out the foot of the perpendicular from a given point to a line.
- Used to set out right angles at a given point on a line in the field.
- Two mirrors may be replaced by two prisms.



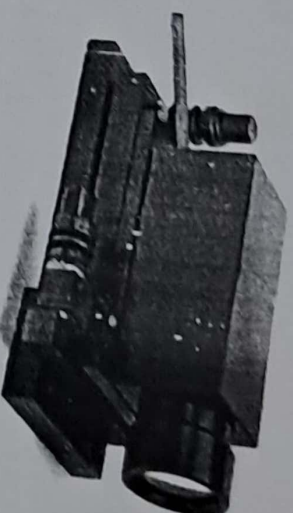
- Visual autocollimators are used for lining up laser rod ends and checking the face parallelism of optical windows and wedges.
- Electronic and digital autocollimators are used as angle measurement standards, for monitoring angular movement over long periods of time and for checking angular position repeatability in mechanical systems.
- Servo autocollimators are specialized compact forms of electronic autocollimators that are used in high speed servo feedback loops for stable platform applications.



An Optical Square

McLeod gauge

- Used to measure vacuum by application of the principle of Boyle's law.
- Works on the principle, "Compression of known volume of low pressure gas to higher pressure and measuring resulting volume & pressure, one can calculate initial pressure using Boyle's Law equation."
- Pressure of gases containing vapours cannot normally be measured with a McLeod gauge, for the reason that compression will cause condensation.
- A pressure from 0.01 micron to 50 mm Hg can be measured. Generally McLeod gauge is used for calibration purpose.



Autocollimator

Laser Scanning Micrometer

- The LSM features a high scanning rate which allows inspection of small workpiece even if they are fragile, at a high temperature, in motion or vibrating.
- Applications :
 - Measurement of outer dia. And roundness of cylinder.
 - Measurement of thickness of film and sheets.
 - Measurement of spacing of IC chips.
 - Measurement of forms.
 - Measurement of gap between rollers.

